

SIMATS ENGINEERING

ASSESSMENT TOOL 2 (AT2) — COURSE OUTCOME 1 (CO1)

Course Name:	Artificial Intelligence	Course Code:	CSA17
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Duration:	1 Hour	Total Marks:	30 (Weightage: 50%)
Course Outcome:	CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		

Code of Conduct & Student Declaration:

I, **Shaik Reehan (Reg No: 192565071)**, certify that this submission is my original work and that I have adhered to the academic integrity guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of Student: Shaik Reehan

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ANNEXURE A: CONSTRAINT-BASED PROBLEM SOLVING

Question 1: Hospital Doctor Shift Scheduling using Backtracking Search

Problem Statement: A hospital needs to assign 3 doctors (D_1, D_2, D_3) to 3 shifts (Morning, Afternoon, Night) subject to: (i) $D_1 \neq \text{Night}$, (ii) $\text{Shift}(D_2) < \text{Shift}(D_3)$ where Morning < Afternoon < Night, (iii) Only one doctor per shift, (iv) $D_3 \neq \text{Morning}$, (v) All doctors assigned exactly one shift.

1. Problem Formulation as a Constraint Satisfaction Problem (CSP)

- Variables:** Set of doctors: $X = \{D_1, D_2, D_3\}$.
- Domains:** $D(D_1) = D(D_2) = D(D_3) = \{\text{Morning (M)}, \text{Afternoon (A)}, \text{Night (N)}\}$. Chronological ordering: Morning (1) < Afternoon (2) < Night (3).
- Constraints:**
 - C_1 (Unary): $D_1 \neq \text{Night}$ (D_1 cannot work Night shift).
 - C_2 (Binary precedence): $\text{Shift}(D_2) < \text{Shift}(D_3)$ (D_2 must work strictly before D_3).
 - C_3 (Global): $\text{AllDifferent}(D_1, D_2, D_3)$ (Only one doctor per shift).
 - C_4 (Unary): $D_3 \neq \text{Morning}$ (D_3 cannot work Morning shift).
 - C_5 (Completeness): Every doctor must be assigned exactly one shift.

2. Node Consistency & Domain Pre-processing

- Applying unary constraint C_1 ($D_1 \neq N$) gives: $D(D_1) = \{M, A\}$.
- Applying unary constraint C_4 ($D_3 \neq M$) gives: $D(D_3) = \{A, N\}$.
- Applying precedence constraint C_2 ($D_2 < D_3$): Since $D_3 \in \{A, N\}$, D_2 cannot be Night (no shift occurs after Night). Hence, $D(D_2) = \{M, A\}$.

3. Step-by-Step Backtracking Search Trace

Step	Variable Tested	Value Assigned	Constraint Check Status	Search Action Taken
1	D_1	Morning (M)	C_1 ($M \neq N$): Satisfied	Valid assignment. Proceed to D_2 .
2	D_2	Morning (M)	C_3 (AllDiff): Fails, $D_2 = D_1 = M$	Conflict! Prune and backtrack.
3	D_2	Afternoon (A)	C_3 (AllDiff): Satisfied ($A \neq M$)	Valid assignment. Proceed to D_3 .
4	D_3	Afternoon (A)	C_3 (AllDiff): Fails, $D_3 = D_2 = A$	Conflict! Prune and backtrack.
5	D_3	Night (N)	C_3 (AllDiff): Satisfied ($\{M, A, N\}$ distinct) C_4 ($N \neq M$): Satisfied C_2 ($A < N$): Satisfied	Valid Solution 1 Found! $\{D_1: M, D_2: A, D_3: N\}$
6	D_1	Afternoon (A)	C_1 ($A \neq N$): Satisfied	Backtrack to explore branch with $D_1 = A$.
7	D_2	Morning (M)	C_3 (AllDiff): Satisfied ($M \neq A$)	Valid assignment. Proceed to D_3 .

Step	Variable Tested	Value Assigned	Constraint Check Status	Search Action Taken
8	D_3	Afternoon (A)	C_3 (AllDiff): Fails, $D_3 = D_1 = A$	Conflict! Prune and backtrack.
9	D_3	Night (N)	C_3 (AllDiff): Satisfied ($\{A, M, N\}$ distinct) C_4 ($N \neq M$): Satisfied C_2 ($M < N$): Satisfied	Valid Solution 2 Found! $\{D_1: A, D_2: M, D_3: N\}$
10	D_2	Afternoon (A)	C_3 (AllDiff): Fails, $D_2 = D_1 = A$	Conflict! Prune. Search complete.

4. Final Valid Schedules

Schedule	Morning Shift (M)	Afternoon Shift (A)	Night Shift (N)	Status
Schedule 1	Doctor 1 (D_1)	Doctor 2 (D_2)	Doctor 3 (D_3)	VALID
Schedule 2	Doctor 2 (D_2)	Doctor 1 (D_1)	Doctor 3 (D_3)	VALID

Deduction: Backtracking search determines that D_3 must work the Night shift in both valid solutions to satisfy $D_2 < D_3$ and $D_1 \neq \text{Night}$. D_1 and D_2 divide the Morning and Afternoon shifts.

Question 2: Robot in 5×5 Grid Environment with Manhattan Distance Heuristic

Problem Statement: A robot operates on a 5×5 grid from Start $S(1, 1)$ to Goal $G(5, 5)$. Obstacles block cells at $\{(2, 3), (3, 3), (4, 3)\}$. Allowed moves are Up, Down, Left, Right with unit cost $c = 1$. We apply Best-First Search (BFS) guided by the Manhattan Distance heuristic $h((r, c)) = |r - 5| + |c - 5|$ using a Priority Queue.

1. Environment Setup & Grid Coordinate Mapping

- **Grid Representation:** Discrete Cartesian 2D grid indexed by coordinates (*row, col*) from (1, 1) to (5, 5).
- **Start Node:** $S = (1, 1)$ (Top-Left corner) | **Goal Node:** $G = (5, 5)$ (Bottom-Right corner).
- **Obstacle Placement:** A central vertical wall at cells $\{(2, 3), (3, 3), (4, 3)\}$ blocking direct column-3 traversal.
- **Cost Model:** Uniform step cost $c = 1$ per orthogonal movement.
- **Manhattan Distance Heuristic:** $h((r, c)) = |r - 5| + |c - 5|$. At $S(1, 1)$: $h = 8$; at $G(5, 5)$: $h = 0$.

2. Grid Map Layout

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	[S(1,1)]	[(1,2)]	[(1,3)]	[(1,4)]	[(1,5)]
Row 2	[(2,1)]	[(2,2)]	[OBST]	[(2,4)]	[(2,5)]
Row 3	[(3,1)]	[(3,2)]	[OBST]	[(3,4)]	[(3,5)]
Row 4	[(4,1)]	[(4,2)]	[OBST]	[(4,4)]	[(4,5)]
Row 5	[(5,1)]	[(5,2)]	[(5,3)]	[(5,4)]	[G(5,5)]

Legend: [S] = Start (1,1), [G] = Goal (5,5), [OBST] = Blocked Obstacle Cell

3. Search Methodology: Greedy Best-First Search (BFS)

When BFS uses a heuristic with a Priority Queue, it executes **Greedy Best-First Search**, ordering the queue by $h(n)$. Ties are broken by deeper path cost or standard directional order (Right > Down > Left > Up).

4. Step-by-Step Node Expansion & Priority Queue Trace

Iter.	Dequeued Node	Cost g	Heuristic h	Successors Generated (Node, h)	Priority Queue / Frontier (Node, h, g)
0	—	0	8	Initialize start node $S(1, 1)$	[((1,1), h=8, g=0)]
1	(1, 1)	0	8	(1, 2, h=7), (2, 1, h=7)	[((1,2), h=7, g=1), ((2,1), h=7, g=1)]
2	(1, 2)	1	7	(1, 3, h=6), (2, 2, h=6)	[((1,3), h=6, g=2), ((2,2), h=6, g=2), ((2,1), h=7, g=1)]
3	(1, 3)	2	6	(1, 4, h=5) [(2,3) is obstacle blocked]	[((1,4), h=5, g=3), ((2,2), h=6, g=2), ((2,1), h=7, g=1)]
4	(1, 4)	3	5	(1, 5, h=4), (2, 4, h=4)	[((1,5), h=4, g=4), ((2,4), h=4, g=4), ((2,2), h=6, g=2), ...]
5	(1, 5)	4	4	(2, 5, h=3)	[((2,5), h=3, g=5), ((2,4), h=4, g=4), ((2,2), h=6, g=2), ...]

Iter.	Dequeued Node	Cost g	Heuristic h	Successors Generated (Node, h)	Priority Queue / Frontier (Node, h , g)
6	(2, 5)	5	3	(3, 5, $h=2$)	[((3,5), $h=2$, $g=6$), ((2,4), $h=4$, $g=4$), ...]
7	(3, 5)	6	2	(4, 5, $h=1$)	[((4,5), $h=1$, $g=7$), ((2,4), $h=4$, $g=4$), ...]
8	(4, 5)	7	1	(5, 5, $h=0$)	[((5,5), $h=0$, $g=8$), ((2,4), $h=4$, $g=4$), ...]
9	(5, 5)	8	0	Goal reached! Terminate search.	—

5. Optimal Path and Cost Result

- **Optimal Traversal Sequence:** $(1, 1) \rightarrow (1, 2) \rightarrow (1, 3) \rightarrow (1, 4) \rightarrow (1, 5) \rightarrow (2, 5) \rightarrow (3, 5) \rightarrow (4, 5) \rightarrow (5, 5)$
- **Total Optimal Path Cost: 8 units** (8 orthogonal steps of unit cost 1). The Manhattan distance heuristic successfully guides the robot along the shortest unobstructed path around the obstacles without expanding unnecessary nodes.

Question 3: Autonomous Rescue Robot in Disaster-Affected Building

Problem Description: An autonomous rescue robot operates in a partially observable building modeled as a room grid to locate a survivor at Goal G starting from S . Paths are partially blocked, sensing is local, movement costs 1, and risky zones incur an additional cost of 2.

Task (a): Analysis of Constraints and Impact on Robot Decision-Making

Constraint	Operational Nature	Impact on Decision-Making Process
1. Up, Down, Left, Right Movement	Action Space Restriction	Restricts the robot's local branching factor to at most 4 orthogonal transitions, simplifying local path computation.
2. Blocked Cells (Obstacles)	Structural Inaccessibility	Debris and collapsed walls render paths impassable. The robot must verify passage feasibility before advancing.
3. Limited Sensing (Adjacent Cells Only)	Partial Observability	Prevents complete offline path calculation. The robot must operate under uncertainty, sensing and updating its map incrementally.
4. Move Cost = 1, Risky Zone Cost = +2 (Total 3)	Non-Uniform Step Costs	Forces multi-criteria optimization: a 2-step detour through safe cells (cost $1 + 1 = 2$) is preferred over entering 1 risky room (cost 3).
5. Minimize Cost & Ensure Safety	Performance Metric	Eliminates reckless behavior; requires the agent to balance path length against risk exposure to minimize cumulative damage/cost.
6. Dynamic Knowledge Updates	Online Search Scenario	The agent must interleave perception, cognitive map updating, and real-time replanning rather than executing a static precomputed trajectory.

Task (b): Solution Approach using Online Search Strategy

Proposed Method: Online Real-Time Search Agent with Dynamic Incremental Replanning (Real-Time Adaptive A* / Online DFS with Local Sensing).

Step-by-Step Operational Procedure:

1. **Initialization:** Construct an internal topological grid map M . Initialize unobserved rooms with an optimistic assumption (assumed free, cost = 1). Set current location $u = S$.
2. **Sensory Perception:** At current room u , scan all 4 adjacent rooms using onboard proximity and environmental sensors:
 - Detect physical blockages $\rightarrow$ assign cost $c(u, v) = \infty$.
 - Detect hazardous conditions (smoke, fire, structural stress) $\rightarrow$ classify as Risky, assign cost $c(u, v) = 1 + 2 = 3$.
 - Clear rooms $\rightarrow$ assign nominal cost $c(u, v) = 1$.
3. **Knowledge Base Update:** Integrate new percepts into internal map M . Mark current room u as visited.
4. **Action Selection & Heuristic Evaluation:** For all unblocked adjacent neighbors v , evaluate local cost: $f(v) = c(u, v) + h(v, G)$, where $h(v, G)$ is the Manhattan distance to survivor G . Select neighbor $v^* = \operatorname{argmin}_v f(v)$.
5. **Execution & Movement:** Actuate motors to transition to v^* . Accumulate total path cost: $\text{Cost}_{total} \leftarrow \text{Cost}_{total} + c(u, v^*)$.
6. **Dead-End Recovery:** If all adjacent neighbors are blocked, backtrack to the nearest unexhausted frontier cell using recorded visited history.
7. **Goal Test:** If current cell matches G , deploy rescue beacon, stabilize survivor, and terminate search. Otherwise, repeat from Step 2.

Task (c): Demonstration of Core Artificial Intelligence Concepts

- **1. Intelligent Agents:** The robot functions as a **Model-Based Utility Agent**. Sensors (thermal/lidar) update an internal spatial belief map M , while actuators (tracked drive wheels, gripper) execute actions maximizing safety and mission success.
- **2. Rationality:** An agent is rational if it selects actions that maximize expected performance given percept history. It demonstrates rationality via rigorous trade-offs: choosing a 2-step safe detour (cost 2) over a 1-step risky room (cost 3), but taking the risky room if bypass detours exceed 3 steps.
- **3. Environment Type Characterization:**

Property	Classification	Justification
Observability	Partially Observable	Sensors only perceive adjacent rooms; global building state is occluded.
Determinism	Stochastic	Risky zones introduce unpredictable structural instability, debris falls, and noise.
Episodic / Sequential	Sequential	Current navigation decisions directly constrain future position and battery life.
Static / Dynamic	Dynamic	Disaster conditions (spreading fires, collapses) evolve while robot deliberates.
Discrete / Continuous	Discrete Grid	Environment is structured into discrete rooms with 4 cardinal movement choices.
Single / Multi-Agent	Single-Agent	The robot acts alone against the disaster environment to reach survivor G .

Task (d): Justification of Chosen Approach

- **Failure of Offline Search:** Standard offline algorithms (A^* , Dijkstra) assume complete prior map knowledge. In a disaster building with unmapped debris, an offline plan generated at S fails upon hitting the first collapsed corridor.
- **Optimality of Online Replanning:** The Online Search Agent interleaves local observation with action execution. By adopting the principle of *Optimism Under Uncertainty*, it explores promising corridors, dynamically updates path costs, bypasses newly discovered hazards, and guarantees reaching survivor G with minimum total cost if a connected path exists.